UNIT 7 · TOPIC 8

Bioconversion into Alcohols, Esters and Organic Acids

Syllabus topic: Bioconversion into alcohols, methyl and ethyl esters, organic acids

Bioconversion

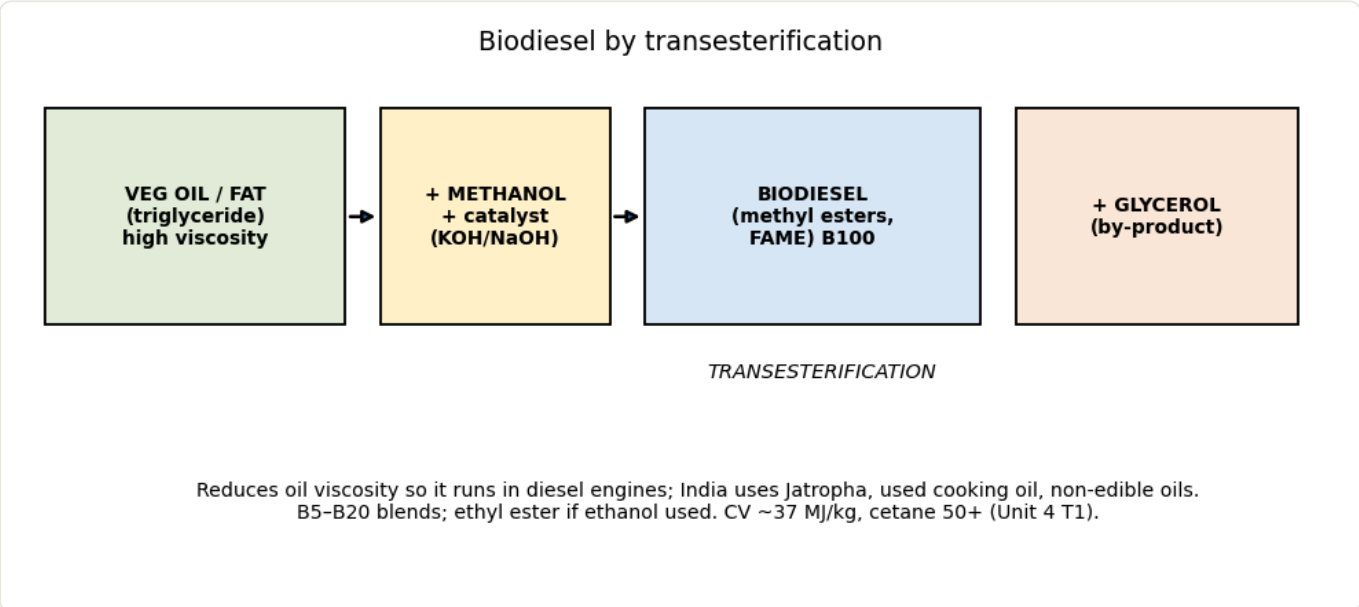
- Bioconversion uses MICRO-ORGANISMS/enzymes to turn biomass into fuels and chemicals — the complement to the thermo-chemical routes (Topics 6–7).
- Wet, high-moisture biomass (dung, sewage, food/agro-processing waste) suits bioconversion; dry biomass suits combustion/gasification.

Alcohols — ethanol

1. Ferment sugars/starch (sugarcane, maize, grains) with yeast.
2. Distil to ETHANOL.
3. Blend with petrol (E20).

Cellulosic (2G) ethanol from straw uses enzymatic hydrolysis first.

Methyl and ethyl esters — biodiesel



Biodiesel by transesterification

Triglyceride (vegetable oil/fat) + methanol (or ethanol) —[KOH/NaOH]→ methyl (or ethyl) esters (biodiesel) + glycerol

- **TRANSESTERIFICATION** with methanol gives methyl esters (**FAME**); with ethanol, ethyl esters. An alkali catalyst (KOH/NaOH) is used.
- It cuts the oil's high viscosity so it runs in diesel engines.
- Feedstocks: Jatropha, used cooking oil, non-edible oils; blends B5–B20.

Organic acids

Microbial fermentation also yields acetic, lactic, citric and butyric acids — platform chemicals from biomass (the biorefinery concept).

Practice MCQs

Q117. Ethanol for blending is produced by:

(a) Transesterification
(b) Fermentation of sugars/starch by yeast
(c) Pyrolysis
(d) Gasification

Q118. Biodiesel is produced by:

(a) Fermentation
(b) Transesterification of oils/fats with an alcohol + catalyst
(c) Distillation of crude
(d) Anaerobic digestion

Q119. Transesterification of vegetable oil with methanol gives biodiesel plus:

(a) Water
(b) Glycerol
(c) Ethanol
(d) CO_2

Q120. The main reason for converting vegetable oil to biodiesel is to:

(a) Increase viscosity
(b) Reduce viscosity so it runs in diesel engines
(c) Add colour
(d) Raise density

Q121. FAME (biodiesel) using ethanol instead of methanol yields:

(a) Methyl esters
(b) Ethyl esters
(c) Glycerol only
(d) Producer gas

Q122. Common alkali catalyst for transesterification is:

(a) H_2SO_4 only
(b) KOH/NaOH
(c) NaCl
(d) $CaCO_3$

Answers

Q117 — (b) Fermentation of sugars/starch by yeast

Q118 — (b) Transesterification of oils/fats with an alcohol + catalyst — *Biodiesel by transesterification (triglyceride + methanol + KOH → methyl esters + glycerol).*

Q119 — (b) Glycerol

Q120 — (b) Reduce viscosity so it runs in diesel engines — *Transesterification reduces viscosity.*

Q121 — (b) Ethyl esters — *Ethanol → ethyl esters.*

Q122 — (b) KOH/NaOH

Sources: Rai — Non-Conventional Energy Sources; Mittal — Energy in Agriculture; MNRE Biogas/SATAT programmes & PIB (as given in the source notes, July 2026); IS 16087:2016 (CBG specification); standard bioenergy texts; NPTEL bioenergy & energy-management lectures.